

GP-NER-20kV

Neutral Earthing Resistor Cabinet





About Us»

GreenPower promotes environmental awareness, and aims to create unprecedented happiness and wealth for our investors, employees, clients and partners. By focusing in the mid-to-high voltage and low voltage eld, and professionally working on R&D, manufacturing, marketing and service of high-end green intelligent switchgears, equipments and products.GreenPower sets to become a well-respected global company in the power industry.

GreenPower, is jointly established by a number of state-owned excellent industrial electrical professional manufacturers, we committed to meet the procurement needs of clients. It is a professional procurement service provider with entity of industrial electrical.

GreenPower provides ONE-STOP solutions service for all customers in the International area. It is located in the time-honored electrical industrial city, a collection of worldclass brand of industrial electrical products,excellent brand of domestic electrical products as the basis.

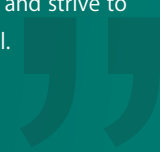
Innovative business philosophy, strong professional supply team, expert technical guidance, advanced Information network management platform, fast logistics, which makes GreenPower as your procurement expert by your side.

GreenPower adheres to its own brand and multi-brand integration, marketing dierentiated development strategy. The products involving low voltage, medium voltage and high voltage transmission and distribution products and industrial automation products.

It covering all aspects of distribution, logistics, warehousing, professional and technical engineering services, systems integration and complete sets of manufacturing. Stable business foundation for cooperation with customers and suppliers, and has established a good reputation to maintain its leading position in the market competition.

Human quests for transcendence, with transcendence we make progress, when the pursuit of unlimited become a belief, our dream began to realize.

Hard-working, dedicated GreenPower people, will be adhering to its past glory, beginners mind, let go sailing, to face the fierce competition in the future, and always help customers make the best choice, and strive to become the best and most reliable procurement service provider in the eld of industrial electrical.



01 Overview

Neutral earthing resistors have been used for more than 100 years in medium- and high-voltage distribution systems in Europe, the United States and other developed countries. In China they are now increasingly applied as well.

When a resonant earthing system using an arc-suppression coil can no longer satisfy the requirements of rapidly developing networks, a neutral point earthing method using a resistor is adopted to: limit system overvoltage, provide a clear and detectable ground-fault current, and allow the faulty feeder to be selected and disconnected quickly.

In generator systems, neutral-earthing resistors are also widely used to limit the fault current at the generator terminals and to control the level of transient overvoltage.

Neutral-earthing resistors effectively limit intermittent arc-grounding overvoltage, reduce operating overvoltage, eliminate resonance overvoltage and make it easy to configure single-phase earth-fault protection which clears the fault in a short time. This helps to reduce the insulation level required for system equipment, extend its service life and improve overall system safety and reliability.

GreenPower has a professional R&D team and advanced manufacturing and test facilities. By introducing and localizing advanced foreign technology, the company focuses on the development and production of neutral-earthing products that effectively reduce overvoltage and improve the reliability and safety of power systems.

We have developed a series of stainless-steel neutral-earthing resistor cabinets for 6-35 kV systems. The resistor elements are made from high-quality imported or domestic stainless-steel alloy, featuring high conductivity, suitable temperature coefficient, excellent corrosion resistance, high temperature capability, high oxidation resistance, high tensile strength and stable resistance value. The products operate safely and reliably.

For medium-voltage systems with delta-connected transformers, a Z-type grounding transformer can be installed to create a neutral point, to which the neutral-earthing resistor is then connected.

02 Service Conditions

- Altitude: ≤ 4000 m
- Ambient temperature: -40°C to $+60^{\circ}\text{C}$
- Relative humidity: $\leq 95\%$ at 25°C
- Installation site: air free from corrosive gases, steam and explosive dust
- Power frequency: 48-52 Hz (50 Hz system), 58-62 Hz (60 Hz system)
- Installation environment: indoor or outdoor
- Neutral displacement under normal conditions: voltage displacement of the neutral point should not exceed 3% of phase-to-earth voltage.

Special service conditions, if any, shall be stated when ordering.

03 Applicable Standards

The design, manufacture and testing of the product can comply with the following standards as required:

- **DL/T 780-2001** Neutral point earthing resistors for distribution systems
- **GB6450** Dry-type power transformers
- **DL/T 620-1997** Overvoltage protection and insulation co-ordination for AC installations
- **GB/T 16927.1-2011** High-voltage test techniques – Part 1: General test requirements
- **GB 1208-2006** Current transformers
- **GB/T 4208-2017** Degrees of protection provided by enclosures (IP code)
- **IEEE Std 32-1972** Requirements, terminology and test procedures for neutral-grounding devices

04 General specifications and technical parameters

6.3kV, 10kV, 35kV transformer neutral earthing resistors parameters

Model	System rated voltage (kV)	Short-time current (A)	Duration of short-time current (s)	Nominal resistance (Ω)	Remark
GP-NER-6/36.4	6.3	100	10	36.40	
GP-NER-6/18.20	6.3	200	10	18.20	
GP-NER-6/9.09	6.3	400	10	9.09	
GP-NER-6/6.06	6.3	600	10	6.06	
GP-NER-10/15.16	10.5	400	10	15.16	
GP-NER-10/10.10	10.5	600	10	10.10	
GP-NER-10/7.57	10.5	800	10	7.57	
GP-NER-10/6.06	10.5	1000	10	6.06	
GP-NER-20/11.54	35	150	10	77	
GP-NER-35/10.10	35	1000	10	20.20	

Notes:

- Resistors can be designed and manufactured on request with resistance values from 1 Ω to 2000 Ω , rated current from 1 A to 2000 A and permissible current duration of 10 s, 15 s, 30 s, 60 s or 2 h. Resistance tolerance: $\pm 3\%$.
- Cabinets can be supplied for indoor or outdoor installation. Enclosures are made of stainless-steel plate or cold-rolled steel with anti-corrosion coating and a high degree of protection.
- The cabinet can be equipped with an isolating switch for easy maintenance, as well as heating and humidity-control devices.
- An intelligent monitoring unit can be installed to measure the neutral unbalanced current, resistor value and cabinet temperature during normal operation. It can also record the magnitude of single-phase earth-fault current and the number of fault operations. A communication interface can be provided so that data and event records can be transmitted to the control room.

05 Main Features

■ Accurate protection

GP-NER neutral-earthing resistor cabinets are used in systems with low-resistance or resistance grounding. In the event of a single-phase earth fault, the faulty feeder is tripped immediately. The resistor provides an additional resistive current so that the total earth-fault current becomes predominantly resistive, limiting the earth-fault overvoltage to within 2.6 times the phase-to-earth voltage.

■ Compact structure and easy installation

The GP-NER cabinet integrates the Z-type grounding transformer (if required), resistor elements, current transformers, measuring instruments, protection terminals and other accessories into a single metal enclosure. Isolating switch and surge arresters can also be included. The layout is clear and neat, installation is convenient and operation and maintenance are safe and reliable.

■ High-quality components

The grounding transformer is a dry-type transformer with a "Z" connection on the primary side. The resistor elements are made from stainless-steel nickel-chromium alloy with high conductivity and current-carrying capacity. They can withstand high temperature (up to 1600°C), have a temperature coefficient not worse than $-0.045\%/^{\circ}\text{C}$, offer stable resistance, good corrosion resistance and explosion-proof performance and provide high reliability. The alloy resistor components are assembled into resistor units, with multiple units welded into a frame structure using argon-arc welding. The resistor units are supported by high-temperature insulating material (polymer). Imported resistor elements can be supplied on request.

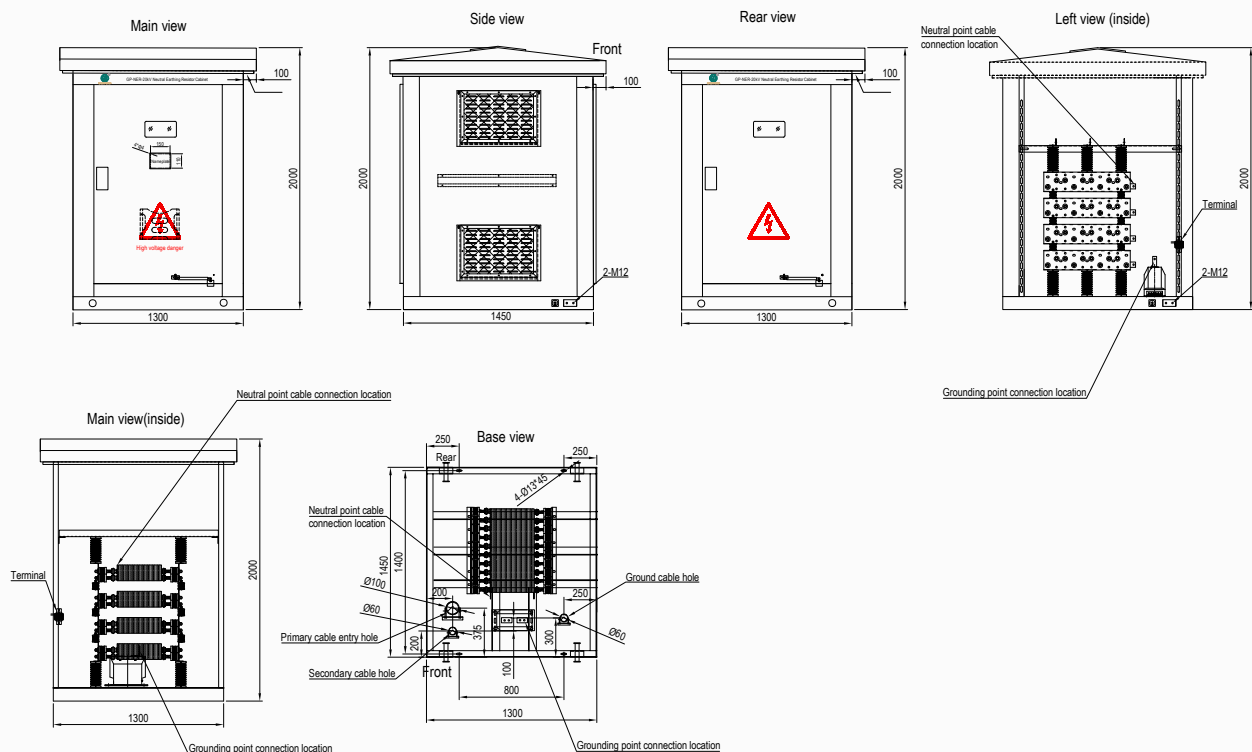
■ Complete monitoring and outputs

The intelligent monitoring device (if provided) continuously monitors neutral unbalanced current, resistance value and internal temperature of the cabinet. It records the instantaneous current during a single-phase earth fault and the number of fault operations. A communication interface is available to transmit measurement and event information to the main control room, enabling operating personnel to obtain real-time data.

■ Strong technical capability and good service

GreenPower is a specialized manufacturer with strong technical expertise. We provide timely technical support and consultation before purchase, and can offer on-site installation guidance and commissioning assistance according to customer requirements.

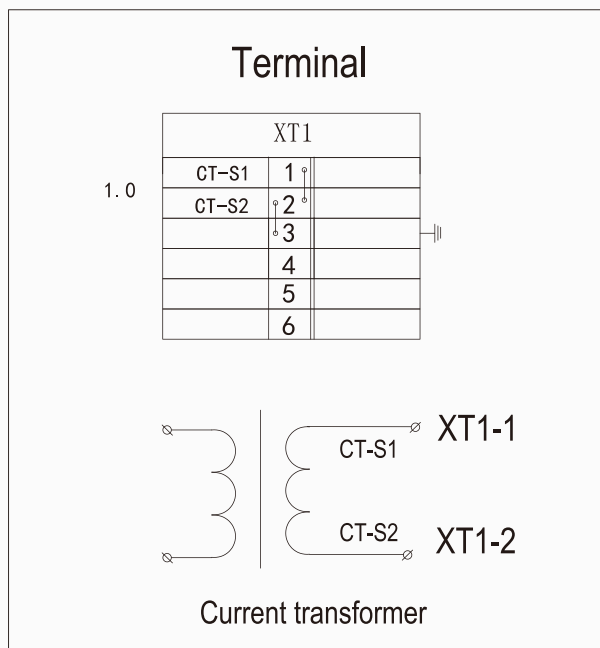
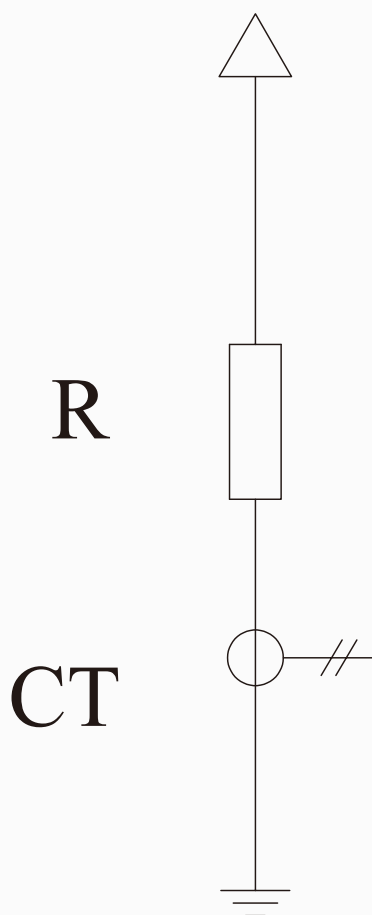
06 Outline dimension



Technical requirements:

1. Cabinet size: 1300*1450*2000mm (width*depth*height)
2. Cabinet material: 2mm Cold-rolled steel plate
3. Cabinet color: RAL7035
4. EEntry and exit methods: cable entry and exit below, with rubber rings and cable clamps for entry and exit holes, fish scale holes for ventilation holes, and dust-proof mesh lining
5. Front and rear doors of the cabinet. The inner side of the door panel and cabinet must have necessary reinforcing ribs
6. Installation method: Outdoor
7. Protection level: IP54
8. Base material: 10 # channel steel. Weld copper grounding blocks on both sides of the channel steel base and process M12 threaded holes.

07 Wiring diagram



No.	Name	Model	Qty.	Remarks
1	Resistor(R)	GP-NER-20kV/77 $\pm 5\%\Omega$	1	150A,10S
2	Current transformer(CT)	LZZBJ9-10 150/1 50Hz	1	Class: 1.0 / 15VA

08 Packaging and Transportation

- The cabinet shall be fixed to the base of the packing case with bolts.
- A plastic film cover shall be provided to protect the enclosure from dust and moisture during transportation and storage.
- The packing case shall be clearly marked with the place of destination, equipment name and transport weight.
- Forklifts or cranes shall be used for unloading. Before unloading, check that the equipment has not been damaged during transport.
- It is recommended to keep the original packing until the equipment has been moved to the installation site, in order to prevent possible damage during handling.

09 Inspection and Acceptance

Upon receipt of the product, the user shall carry out the following checks:

- Check that the contents correspond to the packing list.
- Check that the nameplate data are consistent with the order and contract.
- Check that all factory documents are complete.
- Inspect the enclosure for impact damage or deformation. Check that internal components and connections are intact, tight and free from dust or foreign objects.

If the equipment is not to be put into service immediately after inspection, it shall be stored properly or re-packed.

10 Installation and Pre-commissioning Tests

■ Product installation

The installation, should meet the relevant standards, procedures, the equipment should meet the corresponding voltage level insulation distance requirements, taking into account the equipment of the Ventilation, cooling and equipment installation and maintenance requirements.

Use a forklift to transport the equipment to the installation site. Remove the bolts from the mounting base. Take care to unpack the equipment. Unpack the casing to prevent damage to the casing, top or side-mounted tubing (if any); Quasi-bottom shell at the angle, so as not to damage the equipment floor.

This product can be installed on a dedicated foundation or concrete platform, placed horizontally. Bolt holes (Φ 16mm) on the bottom of the cabinet are fixed on the foundation with anchor bolts. If it is installed on concrete foundation, it is recommended to fix it with expansion bolts.

The earthing terminal of the resistance element is reliably connected to the earthing grid through a single-core cable or a connection line. Cable end of the cable from the bottom up through the bottom of the cable hole into the resistance cabinet, good cable head, and the resistance element earthing or (zero sequence) current transformer output terminal reliable connection cable at the perforation at the use of cable retaining ring Has been provided) protection, and to block. The terminal of the resistance cabinet is connected with a single-core cable or a copper bar to the incoming terminal of the earthing variable resistor or the resistor. The second wiring of the cabinet is checked and tightened all the fixing and connecting bolts in the cabinet to ensure a firm and reliable connection. The shell is grounded reliably by the earthing bolt.

■ Put into operation before the inspection and testing

When the equipment is in place, open the front door with the nameplate, check the cabinet equipment and wiring, and note the following: before charging all packaging materials must be removed from the box to avoid fire; carefully check the insulators, Check all electrical connections to ensure that the connection is secure and reliable; all the fastening bolts are all; put into operation before the test, measuring the resistance of the resistance value of the components, measuring insulation (such as insulation, Resistance, according to the provisions of the standard frequency withstand voltage test, the test voltage is less than 85% of the factory test voltage; restore the front panel of the cabinet, ready to put into operation.

11 Product maintenance

■ Earthing resistance cabinet high reliability, generally do not need special power maintenance, it is recommended to use the station or the main transformer substation maintenance inspection opportunities for maintenance, inspection and maintenance must be strictly enforced safety procedures and safety procedures;

- Disconnect the cabinet from the system;
- Open the door, directly check the box with or without exception;
- Performing a cleaning process;
- Check whether the insulators and casing damage, with a shake table to confirm the porcelain insulation;
- Check the resistance of the integrity of components, measured resistance value, the value should be within 10% error;
- Check the internal connection is reliable;
- Check the tightness of the bolts.

12 Ordering Information

Please inform the following parameters when ordering.

1. The system rated voltage: (kV);
2. Short-term allowable current: (A);
3. The nominal resistance: (Ω);
4. Whether to install the current transformer (CT): CT ratio and capacity, compound error;
5. Into line mode: up and down, down into the next, optional;
6. Short-time allow flow time: (s);
7. The resistance of the cabinet material, cabinet (IP) protection, color and installation location;
8. Whether the need to install the ground record and temperature and humidity control device;
9. Whether to install isolation switch, for maintenance.

E2.1302E-202512

Zhejiang Greenpower Electric Co., Ltd
Zhejiang GreenPower I&E Co., Ltd
Professional Power Equipment Provider

The data and illustrations are not binding.
We reserve the right to make modifications following
technical developments to the products.

Copyright 2025 GreenPower.
All rights reserved.



GreenPower

Contact us



www.sinogp.com



gp@sinogp.com



0577 8550 0968



+86 189 8978 2765



No.827 Wenzhou Avenue, Wenzhou,
Zhejiang, 325011, P. R. China